Zechuan Shi

Ph.D. Candidate in Bioinformatics | zechuas@uci.edu | GitHub: https://github.com/rootze

SUMMARY

Bioinformatics Ph.D. candidate specializing in single-cell multi-omics, regulatory genomics, and AI-driven modeling of neurodegeneration. I develop computational frameworks that integrate transcriptomic, epigenomic, and chromatin-accessibility data with network- and machine-learning-based perturbation models to uncover causal regulators of brain cell-state transitions. My work combines advanced statistical genomics with tool development, including *COMPACT* (network-constrained in-silico perturbation) and *scROAD* (single-cell TF occupancy) to reveal disease-associated regulatory architecture and therapeutic targets in Alzheimer's and related dementias. I aim to build an interdisciplinary research program that unites foundation models with context-specific gene regulatory information to mechanistically understand and predict neurodegenerative processes.

EDUCATION

Ph.D. Candidate in Bioinformatics — Expected Jun 2026

University of California, Irvine, Department of Neurobiology and Behavior

Principal Investigator: Vivek Swarup, Ph.D.

Thesis Committee: Kim Green, Ph.D., Leslie M. Thompson, Ph.D., Marcelo Wood, Ph.D., Jing Zhang, Ph.D.

- **Research Direction:** Bioinformatics, Machine Learning, Genomics, GWAS, Single Cell, Alzheimer's Diseases
- My thesis research develops statistical genomics and machine-learning–based modeling frameworks to interrogate Alzheimer's disease mechanisms and prioritize therapeutic targets.

Master of Science in Biotechnology — Sep 2016 – May 2018

Johns Hopkins University

GPA: 3.92/4.0

- **Concentration:** Bioinformatics, Molecular Targets and Drug Discovery Technologies

SOFTWARE PACKAGES/TOOLS

Developer

compact — Github: compact

- an in silico module perturbation method that unlocks a functional understanding of dynamic gene networks in single-cell data

scROAD — Database: scROAD

- a database that offers single-cell cCREs' transcription factor occupancy information generated from snATAC-seq analysis of human postmortem prefrontal cortex (PFC) tissue for dementia research

ArchRtoSignac — Github: ArchRtoSignac

- an R package that allows easier implementation of both ArchR and Signac scATAC-seq analysis pipelines

MODEL-AD Intellicage Data Processor — App: intellicage-ModelAD

- a web application that processes a broad range of rodent behavioral data for the MODEL-AD consortium

Contributor

hdWGCNA — Github: hdWGCNA

- an R package for performing weighted gene co-expression network analysis (WGCNA) in high dimensional transcriptomics data

FIRST/CO-FIRST AUTHOR PUBLICATIONS

* These authors contributed equally

5. **Shi, Z.**, Das, S., Morabito, et al. (2025). Single-nucleus multi-omics identifies shared and distinct pathways in Pick's and Alzheimer's disease. **Science Advances**. | [Full text] | [code] | [data]

- Dissected transcriptomic and epigenomic variation across Pick's disease and Alzheimer's disease.
- Identified disease-enriched noncoding regions and TF occupancy divergence, revealing cell-type–specific regulatory shifts.
- Discovered a distal human-gained enhancer regulating *UBE3A*, a neuronal E3 ubiquitin ligase linked to synaptic dysfunction and neurodegeneration.
- Developed the interactive TF occupancy database ***scROAD***, integrating Cicero and TOBIAS for single-cell co-accessibility and TF binding analysis across disease and control samples.

4. **Shi, Z.***, Morabito, S.*, et al. (2025). In silico module perturbation analysis unlocks a functional understanding of the dynamic gene networks in single-cell data. (Manuscript in preparation)
   - Developed ***COMPACT***, a network-constrained in-silico perturbation framework for modeling cell-type-specific transcriptomic responses in single-cell data.
3. Das, S.*, **Shi, Z.***, et al. (2025). Region-specific oligodendrocyte states shape the brain microenvironment and modulate Alzheimer's disease pathology. (Manuscript in preparation)
2. Childs, J.*, **Shi, Z.***, et al. (2025). Single nucleus RNA sequencing and spatial transcriptomics of mouse habenula after reinstatement of cocaine self-administration. (Manuscript in preparation)
1. **Shi, Z.***, Das, S.*, Morabito, S., Miyoshi, E., & Swarup, V. (2022). Protocol for single-nucleus ATAC sequencing and bioinformatic analysis in frozen human brain tissue. **STAR Protocols**, 3(3), 101491. | [Full text] | [code]

OTHER CO-AUTHOR PUBLICATIONS

9. Dimas, P., Morabito, S., [et al., including **Shi, Z.**], Geschwind, D., Swarup, V., Franklin, R.. (2025). Dysregulation of transcription networks regulating oligodendrogenesis in age-related decline in CNS remyelination. (Manuscript in review; **Nature Neuroscience**)
8. Fotio, Y., Al Masri, S., **Shi, Z.**, et al. (2025). Metabolic reallocation in spinal cord oligodendrocytes drives chronic pain via neuronal β-amyloid production. (Manuscript in review; **Science Translational Medicine**)
7. Tran, K.M., Kwang, N.E., Butler, C.A., [et al., including **Shi, Z.**], LaFerla, F., ... Green, K.. (2025). APOE Christchurch enhances a disease-associated microglial response to plaque but suppresses response to tau pathology. **Molecular Neurodegeneration**. | [Full text]
6. Miyoshi, E., Morabito, S., Henningfield, C.M. [et al., including **Shi, Z.**]. (2024). Spatial and single-nucleus transcriptomic analysis of genetic and sporadic forms of Alzheimer's Disease. **Nature Genetics**. | [Full text] | [data]
5. Tiwari, V., Prajapati, B., Asare, Y. [et al., including **Shi, Z.**]. (2024). Innate immune training restores pro-reparative myeloid functions for remyelination in the aged central nervous system. **Immunity**, 57(9), 2173–2190.e8. | [Full text]
4. O'Shea, T.M., Ao, Y., Wang, S. [et al., including **Shi, Z.**]. (2024). Derivation and transcriptional reprogramming of border-forming wound repair astrocytes after spinal cord injury or stroke in mice. **Nature Neuroscience**. | [Full text]
3. Garcia-Agudo, L.F., **Shi, Z.**, ..., LaFerla, F., ... Green, K.. (2024). BIN1K358R suppresses glial response to plaques in mouse model of Alzheimer's Disease. **Alzheimer's & dementia**: the journal of the Alzheimer's Association, 20(4), 2922–2942. | [Full text] | [code] | [data]
2. Tran, K.M., Kawauchi, S., Kramár, E.A. [et al., including **Shi, Z.**], LaFerla, F., ... Green, K.. (2023). A Trem2R47H mouse model without cryptic splicing drives age- and disease-dependent tissue damage and synaptic loss in response to plaques. **Molecular Neurodegeneration**. | [Full text]
1. Ma, Z., Flynn, J., Libra, G., & **Shi, Z.** (2018). Elevated CO2 Accelerates Phosphorus Depletion by Common Bean (Phaseolus vulgaris) in Association with Altered Leaf Biochemical Properties. **Pedosphere**, 28(3), 422–429. | [Full text]

PRESENTATION

Oral Session

- "The new era of bioinformatics: How foundation models will rewrite the game in functional genomics". **NeuroBlitz**, Irvine, CA. Nov 2024.
- "Recent advances in Perturb-seq". Single Cell Journal Club, **Daniel Geschwind Lab** at UCLA, Los Angeles, CA. Oct 2024. (**Invited Talk**)
- "In Silico Module Perturbation Analysis unlocks a functional understanding of the dynamic gene networks in single-cell data". **Annual Meeting of American Society of Human Genetics**, Denver, CO. Nov 2024.
- "Single-nucleus open chromatin accessibility landscape of Pick's and Alzheimer's disease". **Alzheimer's Association International Conference**, San Diego, CA. Jul-Aug 2022.

Poster Session

- "In silico network reprogramming of oligodendrocytes reveals therapeutic potential in Alzheimer's disease". **Annual Meeting of American Society of Human Genetics**, Boston MA. Oct 2025. (**Reviewers' Choice Abstract Award**)
- "Single-nucleus multi-ome profiling of epigenomic modifications and transcriptome characterization of heterogeneity in early and late-stage Alzheimer's disease". **Alzheimer's Association International Conference**, Philadelphia, PA. Jul-Aug 2024.
- "Epigenetic dysregulation in Pick's and Alzheimer's disease". **15th Annual Emerging Scientists Symposium**, Irvine, CA. Mar 2024.
- "Characterizing BIN1K358R SNP rs138047593 effects in 5xFAD plaque pathology of Alzheimer's disease using snRNA-seq". **Annual Meeting of American Society of Human Genetics**, Washington DC. Nov 2023.

Fellowship & Awards

ASHG Reviewers' Choice Abstract Award (2025): Awarded to the top 10% of submissions as scored by reviewers

Biological Sciences Travel Award (2023, 2024, 2025): University of California Conference Travel Award ($900)

Bio2X Outstanding Trainee Award (2021-2022): BlueRun Ventures Winter 2021 - 2022 Bio2X Life Sciences Ideation Fellowship

Cum Laude (2015): Graduating with Honor, TSU

Gerhardt Research Fellow (2015): a competitive research award ($3,000)

Grants-in-Aid of Scholarship and Research (GIASR) (2014): a competitive research grant for the initial stage of an individual student research project ($750)

President's Honorary Scholarship (2011-2015): an annual academic achievement scholarship, TSU ($18,000)

INDUSTRY & TRANSLATIONAL EXPERIENCE

Biotech Investment Fellow, BlueRun Ventures – Beijing, China (Remote) — Nov 2021 – Jan 2022

- Participated in a selective Bio2X venture program focused on biotechnology innovation and commercialization strategy
- Delivered bi-monthly reports on biotech industry trends, investment landscapes, and startup evaluations
- Collaborated with venture analysts to conduct due diligence and generate scientific insight for early-stage investment decisions
- Awarded for exceptional performance in biotechnology innovation analysis and startup evaluation

Preclinical Research Specialist, University of Pennsylvania – Philadelphia, PA — Jul 2018 – Jul 2020

Advisor: Lewis Chodosh, M.D., Ph.D.

Project: Recurrent Breast Cancer Preclinical Drug Development

- Led preclinical oncology studies using MTB/TAN bi-transgenic HER2+ mouse models to evaluate therapeutic response and tumor recurrence
- Managed in vivo studies involving >600 mice across 11 transgenic lines to support target validation and efficacy profiling
- Performed RNA-Seq analysis to characterize minimal residual disease and identify prognostic biomarkers in

disseminated tumor cells

- Collaborated cross-functionally with molecular biologists and pharmacologists to integrate transcriptomic data with phenotypic drug response

Consultant, PBG Healthcare Consulting – Philadelphia, PA Sep 2018 – Dec 2018
Wharton School of the University of Pennsylvania

- Collaborated with data scientists and medical researchers to optimize drug development pipelines
- Conducted market research and due diligence in partnership with business teams to assess therapeutic positioning
- Designed and distributed a primary research survey, identified key opinion leaders, and conducted interviews to define unmet clinical needs

R&D Intern, Gracomics LLC (Startup), MO Jan 2016 – May 2016
Advisor: Wayne Zhou, Ph.D.
Project: ELISA-based Diagnostic Kit Development

- Assisted the in-cell ELISA-based diagnostic kit development and assay optimization

EARLY RESEARCH EXPERIENCE

Summer Research Fellow, University of Pennsylvania – Philadelphia, PA May 2013 - August 2013
Advisor: Andy Minn, MD., Ph.D.
Project: Stromal Regulation of Breast Cancer DNA Damage Resistance

- Elucidated a DNA damage resistance mechanism against radiation by activating STAT1 and NOTCH3 due to the interaction between stromal cells and BrCa cells
- Performed tissue culture, siRNA knockdown, flow cytometry, RNA extraction and Q-RT PCR
- Tested Interferon antibody efficiency and analyzed the optimal range of interferon concentration

TEACHING & OUTREACH EXPERIENCE

Program Manager, genPALs and UCI GPS – STEM Data Science Cohort – Irvine, CA Jul 2022 – Jun 2024

- Organized bi-weekly seminars to promote the research in genomics and epigenomics
- Foster connections with invited speakers and industry professionals
- Coordinate with board members to organize professional networking events

Graduate Teaching Assistant, University of California – Irvine, CA Mar 2022 – Mar 2024
Classes: Bio 38: Mind, Memory, and Brain; Bio 48: The Mind-Body Connection in the Neuroscience of Well-Being; N 138: Sex Influences on Brain; NEURBIO 227. Bioinformatics and Systems Biology

- Graded 200+ students' writing assignments, programming tasks and exams
- Facilitated in-class discussion sections and answered students' questions